

Vydhra Python + AI Job-Ready Bootcamp

Duration: 8 Weeks

Format: 2 Topic Points + Practice

Level: Beginner to Job-Ready Foundation

Website-Style Course Structure

Module 01	Week 1 - Introduction to Python	
Topics Covered	Python overview, career paths, installation, VS Code/Jupyter setup	
	Variables, data types, input/output, operators, and first mini programs	
Practice	Setup checklist, calculator, and input-output exercises	
Outcome	Students can install Python, understand syntax, and write their first small programs confidently.	

Module 02	Week 2 - Control Flow	
Topics Covered	If/elif/else, comparison operators, logical operators, and nested conditions	
	Loops: for, while, range, break/continue, and beginner logic-building tasks	
Practice	Grade checker, number logic, and a menu-driven practice task	
Outcome	Students can build decision-based and repeated-execution programs with clean logic.	

Module 03	Week 3 - Functions & Debugging	
Topics Covered	Functions, parameters, return values, scope, and reusable code design	
	Built-in functions, debugging basics, error tracing, and practice patterns	
Practice	Calculator with functions, converter, and reusable utility tasks	
Outcome	Students can organize code into functions and debug basic programming issues.	

Module 04	Week 4 - Data Structures & Strings	■
Topics Covered	Strings, indexing, slicing, common methods, and text handling	
	Lists, tuples, sets, dictionaries, and looping through collections	
Practice	Contact book, shopping list manager, and word counter	
Outcome	Students can store, access, and manipulate structured data effectively.	

Module 05	Week 5 - Files, Errors & Modules	■
Topics Covered	File handling, reading/writing text files, and CSV/JSON basics	
	try/except/finally, importing modules, packages, pip, and project structure	
Practice	Notes app, CSV reader, and an error-safe script	
Outcome	Students can create practical scripts that handle data and runtime errors.	

Module 06	Week 6 - OOP & GitHub Workflow	■
Topics Covered	Classes, objects, constructors, methods, and basic OOP thinking	
	Git basics, GitHub repositories, commits, push workflow, and portfolio hygiene	
Practice	Student record class and publishing the first project to GitHub	
Outcome	Students can build basic class-based programs and publish work professionally.	

Module 07	Week 7 - APIs, SQL & AI Basics	■
Topics Covered	API basics with requests, JSON responses, and mini integration workflow	
	SQL basics, prompt engineering, and intro to LLM/API-based AI workflow	
Practice	Weather or currency app, simple SQL tasks, and prompt design practice	
Outcome	Students gain beginner exposure to data querying, APIs, and modern AI workflow concepts.	

Module 08	Week 8 - Capstone & Job Readiness	■
Topics Covered	Capstone planning, development workflow, testing, debugging, and cleanup	
	Resume, LinkedIn, GitHub portfolio polish, and Python interview preparation	
Practice	Final capstone, README, resume draft, and mock interview practice	
Outcome	Students finish with a portfolio-ready project and stronger fresher-level job readiness.	